

Name: \_\_\_\_\_

Date: \_\_\_\_\_

**Problem 1.**

Convert each decimal number to hexadecimal:

26 = \_\_\_\_\_

60 = \_\_\_\_\_

175 = \_\_\_\_\_

255 = \_\_\_\_\_

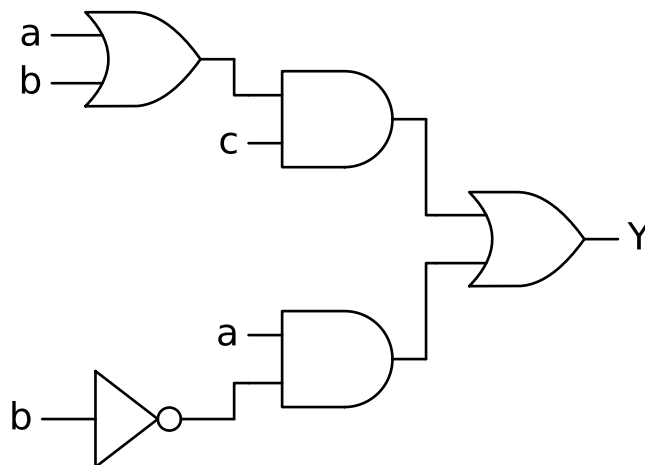
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**Problem 2.**

Convert each hexadecimal number to decimal:

 $1A_{16}$  = \_\_\_\_\_ $2F_{16}$  = \_\_\_\_\_ $B4_{16}$  = \_\_\_\_\_ $FF_{16}$  = \_\_\_\_\_

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**Problem 3.**

Write the Boolean expression for output  $Y$ , then draw its truth table.

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#### Problem 4.

Convert each binary number to hexadecimal:

$$11010_2 = \underline{\hspace{2cm}}$$

$$101111_2 = \underline{\hspace{2cm}}$$

$$10110100_2 = \underline{\hspace{2cm}}$$

$$1111_2 = \underline{\hspace{2cm}}$$

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#### Problem 5.

Convert each hexadecimal number to binary:

$$3C_{16} = \underline{\hspace{2cm}}$$

$$A7_{16} = \underline{\hspace{2cm}}$$

$$1E_{16} = \underline{\hspace{2cm}}$$

$$FF_{16} = \underline{\hspace{2cm}}$$

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#### Problem 6.

Every device on a network has an *IP address*: four numbers separated by dots, each stored in one byte (so 0 to 255). Convert each part of these addresses to 8-bit binary.

$$192.168.1.1 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

$$10.0.0.255 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

$$8.8.8.8 = \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}} \cdot \underline{\hspace{1cm}}$$

Which one of these could *not* be a real IP address, and why?

192.168.0.1      10.0.256.4      172.16.0.255

*Fun fact: 8.8.8.8 is one of Google's real public addresses.*

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### Problem 7.

A black-and-white picture is a grid of *pixels*: each pixel is one bit — 1 means a filled (black) square, 0 means blank. Each row of 8 pixels is one byte, given below in hexadecimal. Convert each byte to 8-bit binary, then shade a square wherever there is a 1.

66<sub>16</sub> = \_\_\_\_\_

FF<sub>16</sub> = \_\_\_\_\_

FF<sub>16</sub> = \_\_\_\_\_

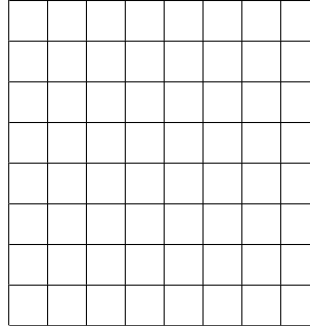
FF<sub>16</sub> = \_\_\_\_\_

7E<sub>16</sub> = \_\_\_\_\_

3C<sub>16</sub> = \_\_\_\_\_

18<sub>16</sub> = \_\_\_\_\_

00<sub>16</sub> = \_\_\_\_\_

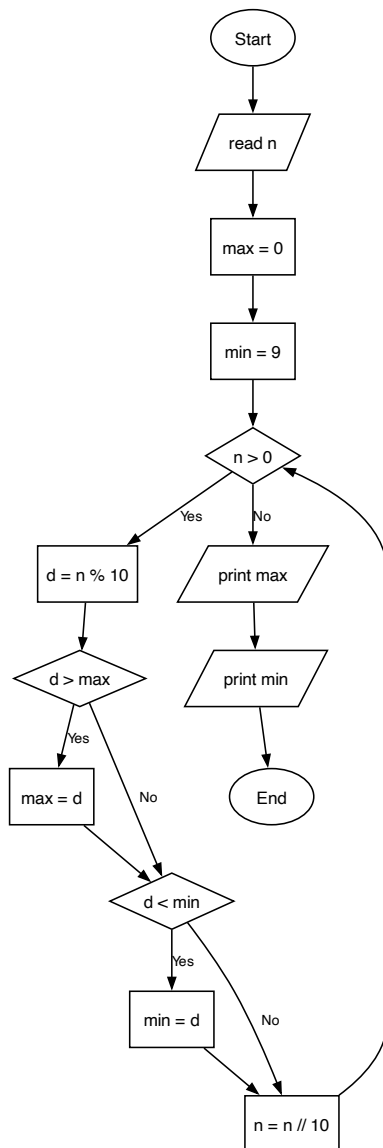


*What picture appears? The whole image is just 8 bytes — 64 ones and zeros. Roughly how many bytes would a picture twice as wide and twice as tall ( $16 \times 16$ ) need?*

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### Problem 8.

Trace this algorithm for  $n = 5184$ , and complete the trace table.

[illegible]